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ORIGINAL RESEARCH



Detection of mobile colistin-resistance gene variants (*mcr-1* and *mcr-2*) in urinary tract pathogens in Bangladesh: the last resort of infectious disease management colistin efficacy is under threat

Bayasrin Ara ^a, Umme Laila Urmi ^a, Tanjum Ara Haque ^a, Shamsun Nahar ^a, Adity Rumnaz ^a, Tamanna Ali ^a, Mohammed Shah Alam ^a, Abu Syed Md. Mosaddek ^b, nor Azlina a Rahman ^c, Mainul Haque ^d and Salequ Islam ^a

^aDepartment of Microbiology, Jahangirnagar University, Dhaka, Bangladesh; ^bDepartment of Pharmacology, Uttara Adhunik Medical College, Dhaka, Bangladesh; ^cDepartment of Physical Rehabilitation Sciences, Kulliyah of Allied Health Sciences, International Islamic University Malaysia, Kuantan, Malaysia; ^dFaculty of Medicine and Defence Health, Universiti Pertahanan Nasional Malaysia (National Defence University of Malaysia), Kuala Lumpur, Malaysia

ABSTRACT

Background: Currently, colistin-resistant pathogens emerged has become a global health concern. This study assessed the distribution of *mcr-1* to *mcr-5* variants with the phenotypic colistin-resistance in bacterial isolates from urinary tract infection (UTI) patients in Bangladesh.

Methods: A cross-sectional study was conducted between April 2017 and March 2018 to enroll uncomplicated UTI patients, and 142 urine samples were analyzed. Uropathogens were identified using the API-20E biochemical panel and 16S rRNA gene sequencing. Polymerase chain reactions detected the *mcr* gene variants in the UTI isolates. The phenotypic colistin-susceptibility was determined by the Kirby–Bauer disc-diffusion method and the minimal inhibitory concentration (MIC) measurement.

Results: The combined carriage of *mcr-1* and *mcr-2* genes in 11.4% (14/123) of urinary tract pathogens. The *mcr*-positive pathogens include five *Escherichia coli*, three *Klebsiella pneumoniae*, three *Pseudomonas putida*, two *Enterobacter cloacae*, and one *Enterobacter hormaechei*. The *mcr*-positive variant showed significantly higher phenotypic colistin resistance with MIC between >16 µg/mL and >128 µg/mL ($p < 0.001$). Over 85% of colistin-resistant isolates showed MDR phenomena.

Conclusions: The emergence of the clinical MDR pathogens with resistance to a highly selective drug may lead to a lack of treatment options for the infectious diseases and spread of infection to the unaffected cohorts.

ARTICLE HISTORY

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KEYWORDS

Mobile colistin-resistance;
human-*mcr*; urinary tract
infections; Bangladesh; MDR

1. Background

Polypeptide antibiotic colistin (Polymyxin E) is considered a last-resort reserved drug for treating severe acute human infections by multidrug-resistant Gram-negative bacteria (GNB) [1]. The drug has been used worldwide in feeding animals for decades for therapeutic and growth promotion purposes [2,3]. Colistin has been banned in some countries as an animal-feed growth promoter to boost farm animals' weight gains [4,5]. However, in many Asian countries, including Bangladesh, the indiscriminate use of colistin in animal husbandry generates selection pressure to develop antimicrobial resistance (AMR) in pathogens and other diverse commensal microbial populations [6,7].

Colistin-resistant pathogens were reported to cause substantial case fatality and have become an increasing global health concern in recent years [2]. Earlier studies have reported the genotypic linkage of the mobile colistin

resistance genes, *mcr-1*, to the phenotypic colistin-resistance [4,8]. Acquisition of colistin resistance by a novel plasmid-mediated gene, *mcr-1*, was first described in Enterobacteriales from both farm-animal products and humans [2]. Afterward, the variants of *mcr*-carrying multiple species of the Enterobacteriales were detected in many countries from environments, animals, and humans [2,9–12]. A substantial abundance of colistin-resistant Enterobacteriales from poultry-chicken gut content was identified in Bangladesh [13]. Usually, a substantial part of Enterobacteriales is normal flora. However, a few of those microbial pathogens can cause systemic bacteremia, community-acquired infections, healthcare-associated infections (HAI), and complicated and uncomplicated urinary tract infections (UTIs) [14–16]. The presence of *mcr-1* has been detected worldwide in the last two years (2018–2019) in MDR Enterobacteriales from human infection, including UTI [17–20]. To the best of our knowledge, the emergence of *mcr*-gene

CONTACT Dr. Md. Salequ Islam  salequ@juniv.edu  Department of Microbiology, Jahangirnagar University, Savar, Dhaka -1342, Bangladesh, Telephone: +880-1715-029136, Fax, +880-2-7791052.; Professor Dr. Mainul Haque  runurono@gmail.com; mainul@upnm.edu  Faculty of Medicine and Defence Health, Universiti Pertahanan Nasional Malaysia (National Defence University of Malaysia), Kem Sungai Besi, 57000 Kuala Lumpur, Malaysia. my Cell Phone: +60109265543

 Supplemental data for this article can be accessed [here](#).

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Article highlights

- The mobile colistin resistance gene variants, *mcr-1* and *mcr-2*, were detected in UTI isolates of Bangladesh.
- *Escherichia coli*, *Klebsiella pneumoniae*, *Enterobacter cloacae*, and *Pseudomonas putida* are the prominent UTI isolates reported of the *mcr* carriage.
- Uropathogens carrying the *mcr* variants showed significantly higher phenotypic colistin resistance with MIC values for colistin-sulfate ranging between >16 $\mu\text{g/mL}$ and >128 $\mu\text{g/mL}$.
- Females were diagnosed with UTI more commonly than males. However, *mcr* genes were identified as a higher proportion in males compared to females.
- Among the *mcr*-positive uropathogens, 85% were found MDR that poses severe threats to colistin's clinical utility, a last resort reserve drug.

variants linked to colistin-resistant in human urinary tract infections in Bangladesh is not studied in-depth and in detail [21]. This study was designed to investigate different variants of the *mcr*-gene and their association to colistin-resistance in UTI bacterial pathogens in Bangladesh and the demographic factors associated with the identified *mcr* variants.

2. Materials and methods

2.1. Study design and sampling

A cross-sectional study was conducted between April 2017 and March 2018 to enroll uncomplicated UTI patients who attended outpatient Departments at Gonoshasthaya Medical College Hospital and Enam Medical College and Hospital in Savar, Dhaka, Bangladesh. An uncomplicated UTI was defined as urinary infections with no structural or functional abnormalities and when the infected persons are not pregnant or have not been instrumented (i.e. catheter-assisted) [22]. In the previous two weeks, selected patients who had not taken antibiotic treatment were requested to participate in the study. To ensure uncomplicated UTI, critically ill and immunocompromised patients, mainly those who have cancer, organ transplant, renal disorders, tuberculosis, and HIV/AIDS, were excluded from the study. We only included culture-positive cases in our analysis. Further, the exclusion criterion was set for the UTI cases that yield two or more different bacteria from a single urine specimen. Informed written consent was obtained from each adult study participant before collecting their demographic data and urine specimens. For minor participants, consent was obtained from their parents or legal guardians. Associated clinical signs and symptoms of the patients were documented using a structured questionnaire. The participants have guided the standard procedure of collecting clean-catch midstream urine. The samples were collected accordingly in sterile glass tubes and transferred to the laboratory for further analysis.

2.2. Bacterial isolation and identification

Collected urine samples were inoculated on MacConkey agar (Liofilchem, Italy) and cysteine-, lactose-, and electrolyte-deficient (CLED) agar (Liofilchem, Italy) simultaneously.

MacConkey agar supports gram-negative UTI pathogens (Supplementary Figure 1A), while CLED agar aids in the growth of gram-negative bacteria and gram-positive cocci if present in urine samples. Urine cultures were incubated at 37°C overnight in aerobic conditions. Colony counts of 10^2 or 10^3 CFU/mL were considered a cutoff value for a probable UTI infection [23]. Gram's staining and biochemical tests initially identified Growth-positive bacteria. A rapid biochemical-test kit API 20E (BioMe'rieux, Durham, NC), consisting of carbohydrate batteries and enzymatic substrates in a set of chromogenic panels, was used to confirm the isolated identity (Supplementary Figure 1B) [24]. A part of the bacterial identity was validated by the polymerase chain reaction (PCR) amplification and sequencing of the 16S rDNA gene [25]. The isolates were preserved in 30% glycerol in Trypticase Soy Broth (TSB) at -80°C until further antimicrobial susceptibility analysis.

2.3. Phenotypic colistin susceptibility testing

The phenotypic susceptibilities of antibiotics, including colistin, of the UTI isolates, were tested primarily by the disc diffusion method (Kirby–Bauer) on Mueller–Hinton agar (MHA, Oxoid, Basingstoke, UK) plates according to the European Committee on Antimicrobial Susceptibility Testing (EUCAST) and Clinical & Laboratory Standards Institute (CLSI) guidelines (Supplementary Figure 1 C) [26,27]. Further, the isolates' phenotypic antimicrobial susceptibilities were tested by the agar dilution method [28]. For disc-diffusion, a 3-hour bacterial suspension in Mueller–Hinton broth was prepared to a concentration of McFarland 0.5 equivalent and then streaked on MHA plates using a cotton swab to ensure consistent growth. Discs of antibiotics (Oxoid, Basingstoke, UK), namely, colistin (25 μg), Amoxycillin+Clavulanic acid (30 μg), Cephalexin (30 μg), Cefuroxime Sodium (30 μg), Cefixime (30 μg), Ceftriaxone (30 μg), Cefepime (30 μg), Imipenem (10 μg), Meropenem (10 μg), Doripenem (10 μg), Nalidixic acid (10 μg), Ciprofloxacin (5 μg), Lomefloxacin (10 μg), Ofloxacin (5 μg), Gentamicin (30 μg), Amikacin (30 μg), and Netilmicin (30 μg) were placed on the bacterial lawns and incubated at 37°C overnight to determine their susceptibilities. Sensitive bacteria developed a clear zone around the disc, and zone diameter was measured and evaluated. Bacteria were classified as susceptible (S), intermediate (I), and resistant (R) from the CLSI guideline charts for the appropriate antibiotics tested [27].

The minimal inhibitory concentration (MIC) measurement by agar dilution method further determined the lowest colistin concentration to inhibit the visible growth of bacterial test isolates [28]. For agar dilution MIC, different concentrations of colistin-sulfate powder (Santa Cruz Biotechnology Inc, TX) were incorporated into an MHA medium from 1.0 $\mu\text{g/mL}$ to 128.0 $\mu\text{g/mL}$ in a two-fold dilution order. Each test inoculum was prepared by inoculating one pure culture colony into Mueller-Hinton broth and incubated for three-hour at 37°C that develops a density of inoculum equivalent to 10^4 colony-forming units (CFU) per spot to drop on the MHA. The inoculum density was periodically compared to a 0.5 McFarland standard, equivalent to approximately 10^8 CFU/mL. The plates were incubated at 37°C in the air for 18–20 hours. Agar

dilution MICs were performed in duplicates. The epsilometer test (E-test) was performed parallel partly using a commercial strip containing a predefined gradient of colistin concentrations (Liofilchem Inc, Italy) to validate colistin MIC determination by the agar dilution method (Supplementary Figure 1D) [29,30]. *Escherichia coli* ATCC25922 was used as the susceptible-control reference strain for disc diffusion and MIC testing. Besides, a control plate without colistin-sulfate was examined for the growth of both test and control isolates. The clinical breakpoints for colistin resistance were defined according to the EUCAST and CLSI statements when the MIC value was $>2 \mu\text{g/mL}$ [26].

2.4. Detection of the colistin resistance *mcr* Genes

The isolates were subjected to a singleplex polymerase chain reaction (PCR) to detect the *mcr-1* gene using primers described elsewhere [2] and confirmed by sequencing. Amplicons were visualized under UV light after 1.2% agarose gel electrophoresis. Then, a multiplex PCR was conducted separately to amplify the *mcr-1* to *mcr-5* genes in the isolates. The primer sets for the *mcr-2*, *mcr-3*, *mcr-4*, and *mcr-5* gene variants were obtained from a recently published original study [8]. Briefly, the modified protocol was as follows: 2.0 μL bacterial template DNA was added to a 15 μL 2X PCR premixture (GeneON, Germany), 1 μL each primer (5 pmol/ μL), and the remaining deionized water was added to obtain a final volume of 30 μL . Reactions went through an initial denaturation at 94°C for 15 min followed by 25 cycles of amplification (Applied Biosystems 2720 Thermal Cycler, Singapore), comprising denaturation for 30 s at 94°C, annealing for 90 s at 55°C, and extension for 1 min at 72°C, and a terminal 10 min extension at 72°C. Expected amplified products for *mcr-1* (309 bp), *mcr-2* (715 bp), *mcr-3* (929 bp), *mcr-4* (1116 bp), and *mcr-5* (1644 bp) underwent electrophoresis through 1.2% agarose gel followed by staining with ethidium bromide and were visualized under UV light.

2.5. Statistical analysis

Verified data were entered and subsequently analyzed using IBM SPSS statistics data editor (version 21). Missing data were excluded from the bivariate analysis. Both descriptive and inferential statistical procedures were used to describe the carriage of the *mcr* gene variants in UTI pathogens and their phenotypic attributes. Pearson's chi-square test was used to test any association between categorical data, and Yate's correction for continuity was applied where necessary. If the chi-square test's expected frequency cannot be assumed, the results of Fisher's Exact test of 2×2 contingency table were reported instead. Two-tailed *p*-values were calculated to determine statistical significance, with the significance level was set at 0.05.

2.6. Ethics statements

The Ethics and Research Review Committee of the Jahangirnagar University (JU) Faculty of Biological Sciences approved this study [No. BBEC, JU/M 2017 3(4) dated

15.03.2017]. All the study protocols adhered to the Declaration of Helsinki for recruiting human subjects for medical research. Written informed consent was obtained from each adult study patient for collecting the urine samples. Separate written informed consent was taken from parents or legal guardians for patients under 18 years of age. Samples were coded to anonymize the personal identities and other information of the study participants.

3. Results

3.1. Urinary tract-infected study patients

During the one-year study period, urine culture-positive 142 patients attending two hospitals in Dhaka city with symptomatic UTI were examined, and bacterial isolates were identified. All of the participants were acutely symptomatic. Most of them reported frequent abdominal pain, repeated urge to urinate, painful urination, and an incomplete empty feeling in their bladder. The mean age ($\pm$ standard deviation) of the study patients was 35.7 ± 15.6 years. The age of the patients included ranged from 4 to 76 years. Incidence was higher among females, 76.1% ($n = 108$) compared to 23.9% ($n = 34$) in males. Further analysis revealed higher UTI episodes in females, especially in the younger age group, but it was not significant statistically through Fisher's Exact test ($p = 0.680$) as detailed out in Table 1. A 10-year range age classification showed that the 21–30 years age-group combining both genders accounted for the highest 35.2% (50/142) of total infections. The comparison in more detailed age groups is shown in Figure 1(a), where UTI was identified in all age groups from <10 years to 70+ years.

3.2. Phenotypic colistin susceptibility of the urinary tract pathogens

A total of 142 UTI isolates yielded from the study samples. The isolates were classified as 34 *Escherichia coli* (23.9%), 56 *Klebsiella* spp. (39.4%), 14 *Proteus* spp. (9.9%), five *Staphylococcus* spp. (3.5%), 14 *Pseudomonas* spp. (9.9%), 17 *Enterobacter* spp. (12.0%) and one of *Acinetobacter* spp. and *Citrobacter* spp. As *Proteus* and *Staphylococcus* species are naturally resistant to colistin [27,31,32], we excluded 14 *Proteus* spp. and five *Staphylococcus* spp the subsequent colistin-resistance analyses in this study. Thus, in vitro antimicrobial potency of colistin sulfate (25 ug) was tested for the 123 UTI isolates (excluding *Proteus* and *Staphylococcus* species) based on the disc diffusion method following EUCAST and CLSI guidelines. Bacteria were marked colistin-resistant (R) if its 25 μg disc generated ≤ 10 mm diameter zone of inhibition (Supplementary Figure 1 C). Further, the

Table 1. Associations between sex and age-groups among urine culture-positive patients ($n = 142$).

	Male ($n = 34$) Frequency (%)	Female ($n = 108$) Frequency (%)	<i>p</i> -value
Age ≤ 40 years old	22 (22.9)	74 (77.1)	0.680*
Age > 40 years old	12 (26.1)	34 (73.9)	

*Fisher's Exact test

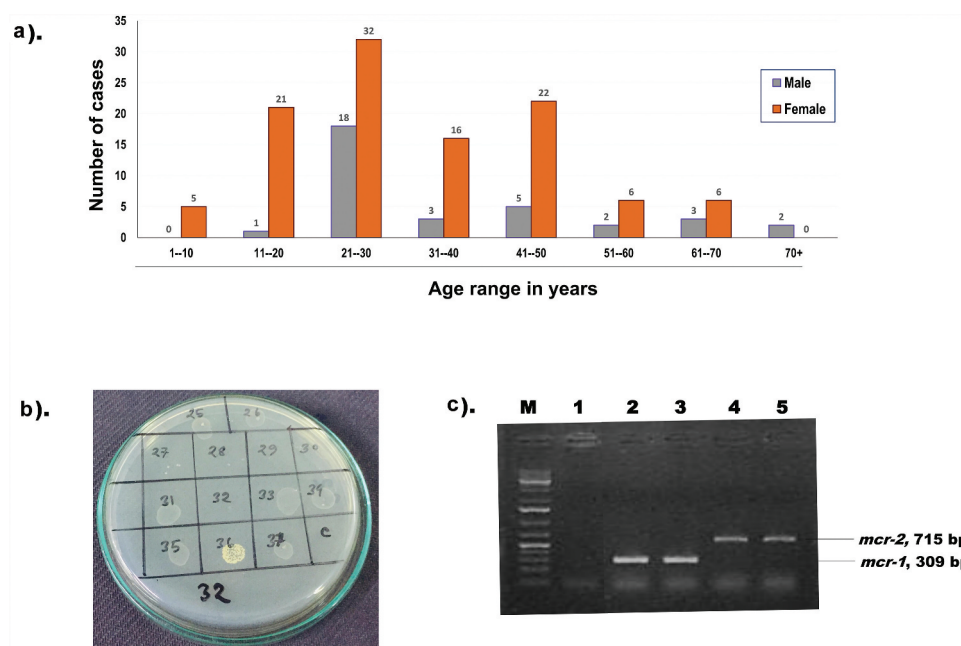


Figure 1. Age and gender-based isolation of urinary tract infecting bacteria, determination of their minimum inhibitory concentration (MIC) for colistin sulphate, and their carriage of the *mcr* genes. (a) Association between sex and age-groups among urine culture-positive patients (n=142). (b). Minimum Inhibitory concentration (MIC) was determination by agar dilution method. The MIC value >2 g/mL was considered resistant, according to the European Committee on Antimicrobial Susceptibility Testing (EUCAST). (c). Colistin resistance gene, *mcr-1* (309 bp) and *mcr-2* (715 bp) were amplified by polymerase chain reaction (PCR). Lanes 2 and 3 showed positive amplification for *mcr-1*; lanes 4 and 5 exhibited for *mcr-2*, and lane-1 showed no amplification in the test templates. Lane M was 1-Kb molecular marker

agar dilution method identified the isolates as susceptible (S) when their growths were absent at ≤ 2 μ g/mL colistin sulfate concentrations and resistant (R) when exhibited growth at >2 μ g/mL. Colistin sulfate showed moderate potency against the test pathogens, where 63 (52.5%) of isolates were resistant by the agar dilution test (Figure 1(b)). A powerful harmony was observed between the phenotypic assessment by agar dilution and disc diffusion methods ($p = .000$). Separately, *E. coli*, *Klebsiella* spp., and *Pseudomonas* spp. showed 48.5%, 44.4%, and 50.0% phenotypic resistance against colistin. In contrast, uropathogen *Enterobacter* spp. exhibited 82.4% resistance. The overall bacterial inter-species analysis for colistin-resistance shows a marginal non-significant difference [$\chi^2(5, N = 123) = 9.54, p = .09$].

3.3. Carriage of the *mcr* gene variants

The colistin resistance genes, *mcr-1* to *mcr-5*, were investigated in 123 UTI isolates using PCR analysis. The *mcr-1* and *mcr-2* genes were detected in 9.8% (12) and 1.6% (2) of isolates, respectively. The combined prevalence of the *mcr* gene variants was 11.4% (14/123). Harborage of the identified *mcr-1* and *mcr-2* was mutually exclusive; no isolate was found with the overlapping carriage of these two *mcr* gene variants (Figure 1(c)). The *mcr* traits were found diversified among the UTI pathogens, including five *Escherichia coli*, three *Klebsiella pneumoniae*, three *Pseudomonas putida*, two *Enterobacter cloacae*, and one *Enterobacter hormaechei* (Table 2). The identified *mcr* genes were higher in males (21.4%, 6/28) compared to females (8%, 8/100) ($p = .07$). Almost uniform

Table 2. Identified urinary tract pathogens carrying *mcr* gene variants^a.

Bacteria types ^b	Number of Isolates carrying <i>mcr</i> genes		Percentage of <i>mcr</i> -positive isolates
	Positive	Negative	
<i>Escherichia coli</i>	5	29	14.7
<i>Klebsiella pneumoniae</i>	3	52	5.5
<i>Pseudomonas putida</i>	3	12	20.0
<i>Enterobacter</i> spp ^c	3	14	17.6
<i>Acinetobacter pittii</i>	0	1	-
<i>Citrobacter braakii</i>	0	1	-
Total	14	109	11.4

^a12 *mcr-1* and two *mcr-2* genes were identified by polymerase chain reaction (PCR).

^bBacteria were primary identified based on their growth on selective culture media followed by biochemical tests. Further, identifications were confirmed by API 20E and 16s rDNA sequencing.

^c*Enterobacter* spp. includes one *Enterobacter hormaechei* and two *Enterobacter cloacae*.

distributions of the genes were identified in different age groups of the study subjects.

3.4. Association of the *mcr* gene variants with phenotypic colistin susceptibilities

All the *mcr*-positive isolates were colistin-resistant by the disc diffusion test. An independent duplicate susceptibility analysis by the agar dilution method generated concordant results (Table 3). Altogether, the isolates possessing the *mcr* variants showed a significant association with the phenotypic colistin-resistance attributes ($P = .000$). Albeit, some *mcr*-negative isolates (42.3%, 52/123) also showed phenotypic colistin-resistance by the broth

Table 3. Phenotypic colistin susceptibility profiles of *mcr*-positive urinary tract infection (UTI) isolates.

<i>mcr</i> -positive isolate ID	Identified bacteria ^a	Identified <i>mcr</i> gene variant	Phenotypic colistin susceptibility by: Disc-diffusion ^b	MIC (μg/mL) ^c
UJ14	<i>Enterobacter cloacae</i>	<i>mcr-1</i>	R	64
UJ32	<i>Pseudomonas putida</i>	<i>mcr-1</i>	R	128
UJ34	<i>Klebsiella pneumoniae</i>	<i>mcr-1</i>	R	64
UJ36	<i>Escherichia coli</i>	<i>mcr-1</i>	R	64
UJ37	<i>Escherichia coli</i>	<i>mcr-1</i>	R	128
UJ47	<i>Enterobacter cloacae</i>	<i>mcr-1</i>	R	32
UJ50	<i>Escherichia coli</i>	<i>mcr-1</i>	R	128
UJ58	<i>Enterobacter hormaechei</i>	<i>mcr-1</i>	R	64
UJ63	<i>Pseudomonas putida</i>	<i>mcr-1</i>	R	32
UJ94	<i>Klebsiella pneumoniae</i>	<i>mcr-1</i>	R	128
UJ98	<i>Escherichia coli</i>	<i>mcr-2</i>	R	16
UJ99	<i>Klebsiella pneumoniae</i>	<i>mcr-2</i>	R	128
UJ107	<i>Escherichia coli</i>	<i>mcr-1</i>	R	64
UJ109	<i>Pseudomonas putida</i>	<i>mcr-1</i>	R	128

^a, Bacteria were identified by rapid biochemical test kit API 20E system (BioMérieux, Durham, NC) followed by 16S rDNA sequencing.

^b, Colistin disc (25 μg, Oxoid, Basingstoke, UK) was used by the disc-diffusion method (Kirby–Bauer) on Mueller–Hinton agar (Oxoid, Basingstoke, UK) plates according to the European Committee on Antimicrobial Susceptibility Testing (EUCAST) guidelines. Sensitive bacteria developed a clear zone around the disc. Bacteria were considered colistin-resistant (R) if its 25μg disc generated ≤10 mm diameter zone of inhibition.

^c, The minimum inhibitory concentration (MIC) measurement was conducted by agar dilution method following the EUCAST guidelines.

dilution test. All the isolates carrying either *mcr-1* or *mcr-2* traits showed high MIC levels of colistin sulfate ranging between

>16 μg/mL and >128 μg/mL (Table 3). An overall invariable result was observed between the agar dilution and the commercial *E-test* methods for colistin MIC determination. Several isolates exhibited mismatching of two- to four-fold MIC values between E-test and agar dilution methods. The majority of *mcr*-naïve isolates demonstrated low MICs of colistin. The alignment of both *mcr*-negative and *mcr*-positive UTI isolates with different colistin MIC levels from ≤1 μg/mL to >128 μg/mL were analyzed. About 53.3% of the *mcr*-negative isolates exhibited MIC at ≤2 μg/mL threshold and the remaining 46.7% aligned with different higher MIC values. However, a reasonable 22.3% of the *mcr*-negative isolates demonstrated MIC as high as >128 μg/mL (Figure 2). None of the *mcr*-positive isolates lay at ≤2 μg/mL threshold; instead, all tied up with higher MIC levels above 16 μg/mL at least. Over 46% of the isolates reside at the highest MIC (>128 μg/mL) edge (Figure 2). The control plate without colistin sulfate supported adequate growth of all the test-isolates. *Escherichia coli* ATCC25922 strains were used as susceptible control that exhibited MIC <2 μg/mL each time.

3.5. Multi-drug resistance and *mcr*-carriage

The MDR phenomena of all the 14 colistin-resistant isolates carrying either *mcr-1* or *mcr-2* were analyzed. The isolated bacteria were then tested for their susceptibilities by the disc-diffusion to other 17 different antibiotics from six drug-classes. MDR was adopted as acquired non-susceptibility to at least one agent in three or more antimicrobial categories, as defined by the Centers for Disease Control and Prevention (CDC) [33]. The analyses revealed over 85% (12/14) of the *mcr*-bearing isolates as MDR (Table 4). Also, 63.3% of *mcr*-negative isolates appeared MDR with similar analyses (data not shown).

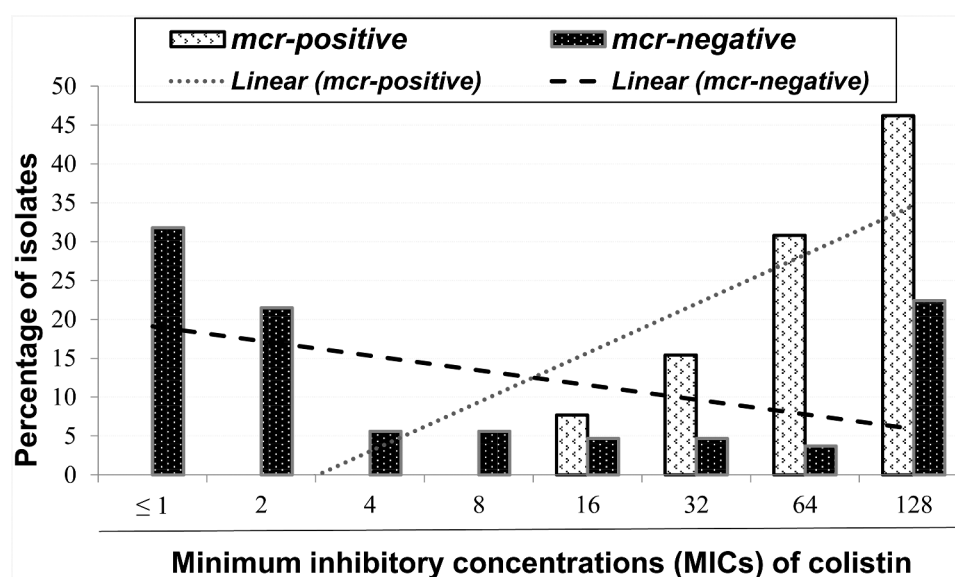


Figure 2. Acquisition of the *mcr*-gene acquisition and minimum inhibitory concentrations (MICs) of the identified UTI isolates (n=128). All the bacteria in this study were subjected to measure minimum inhibitory concentrations (MICs) of colistin between 1.0 μg/mL and 128.0 μg/mL by agar diffusion dilution tests. Percentages of bacteria exhibiting each MIC level were calculated. The X-axis shows each of the MIC points, and Y-axis represents the percentages of isolates at a particular MIC. The percentages of *mcr*-positive and *mcr*-negative isolates at MIC endpoints were compared. White-dot bars show the magnitudes of *mcr*-positive isolates, and the black-dot bars represent *mcr*-negative. Percentile distributions show that *mcr*-positive isolates exhibited higher MICs compared to *mcr*-negative isolates. The upward linear trend line indicates the gradual increase of the *mcr*-positive isolate frequency towards higher colistin MIC levels. The downhill trend line suggests decreasing proportions of the *mcr*-negative uropathogens towards higher colistin MIC values.

Table 4. Multi-drug resistance (MDR) phenomena of *mcr*-positive urinary tract infection (UTI) isolates.

List of antibiotics tested (n = 17, from six drug-classes)		Phenotypic susceptibilities of <i>mcr</i> -positive UTI isolates														
Drug class	Antibiotic Name	UJ14	UJ32	UJ34	UJ36	UJ37	UJ47	UJ50	UJ58	UJ63	UJ94	UJ98	UJ99	UJ107	UJ109	
β-lactam with β-lactamase inhibitor	Amoxi-clav ^a	- ^b	- ^c	R	R	R	- ^b	R	- ^b	- ^c	R	I	R	S	- ^c	
	Cephalexin-G1 ^d	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
Cephalosporins	Cefuroxime-G2	R	S	R	R	R	R	R	R	R	R	R	R	S	R	
	Cefixime-G3	R	R	R	S	R	R	R	R	R	R	R	R	R	R	
	Ceftriaxone-G3	R	R	R	S	R	R	R	R	S	R	R	R	S	R	
	Cefepime-G4	R	R	R	S	R	R	R	R	R	R	I	R	S	R	
Carbapenems	Imipenem	R	R	S	S	I	R	I	R	R	S	S	S	S	I	
	Meropenem	S	S	S	S	S	S	S	R	R	S	S	S	S	S	
	Doripenem	S	S	S	S	S	S	R	S	S	S	S	S	S	S	
Quinolone and fluoroquinolones	Nalidixic acid	R	R	R	R	R	I	R	R	R	R	R	R	R	R	
	Ciprofloxacin	I	R	I	R	R	S	R	I	R	R	S	R	S	R	
	Ofloxacin	R	R	S	R	R	S	R	R	R	R	S	R	S	R	
	Lomefloxacin	R	S	S	R	R	R	R	R	R	R	S	R	S	R	
Aminoglycosides	Gentamicin	R	S	S	S	S	S	S	R	S	S	S	S	S	R	
	Amikacin	I	S	S	I	I	S	I	I	I	R	S	S	S	S	
	Netilmicin	R	S	R	R	R	R	R	R	S	I	S	S	S	R	
Polymyxins	Colistin	R	R	R	R	R	R	R	R	R	R	R	R	R	R	
MDR status ^e		+	+	+	+	+	+	+	+	+	+	-	+	-	+	

R, Resistant; S, Sensitive; I, Intermediate; MDR, multidrug resistant.

^aAmoxicillin-clavulanic acid,

^bUJ14, -47, -58 are *Enterobacter* spp., which is naturally resistant to Amoxicillin-clavulanic acid, therefore, AST results has been excluded from MDR calculation.

^cUJ 32, -63, -109 are *Pseudomonas* spp. which is naturally resistant to Amoxicillin-clavulanic acid, therefore, AST results has been excluded from MDR calculation.

^ddifferent generations of cephalosporin.

^eMultidrug-resistant isolate: defined by center for disease control (CDC) as an isolate that is resistant to at least one antibiotic in three or more drug classes (<https://www.cdc.gov/narms/resources/glossary>). '+', indicates MDR-positive; and '-', indicates MDR-negative.

Table 5. Demographic factors associated with *mcr*^a-carriage (n = 123).

	Frequency (%)	Frequency (%)	p-value
	<i>mcr</i> -1 (+ve) (n = 12)	<i>mcr</i> -1 (-ve) (n = 111)	
Sex:			
Male	4 (33.3)	22 (20.7)	0.294*
Female	8 (66.7)	89 (79.3)	
Age-group:			
≤ 40 years old	10 (83.3)	71 (63.8)	0.217*
> 40 years old	2 (16.7)	40 (36.2)	
	<i>mcr</i> -2 (+ve) (n = 2)	<i>mcr</i> -2 (-ve) (n = 121)	
Sex:			
Male	1 (50.0)	25 (21.4)	0.391*
Female	1 (50.0)	96 (78.6)	
Age-group:			
≤ 40 years old	1 (50.0)	80 (65.9)	1.000*
> 40 years old	1 (50.0)	41 (34.1)	

^amobile colistin resistance, *Fisher's Exact test

3.6. Demographic factors associated with the *mcr*-carriage

Since the expected frequency assumption for the chi-square test was not satisfied for the statistical analysis done even with a 2 × 2 contingency table, Fisher's Exact test results are reported to assess the demographic factors (sex and age groups) associated with *mcr*-carriage identified in this study, namely the *mcr*-1 and *mcr*-2. The results indicated that sex and age-groups were not significantly associated with the presence of *mcr*-1 and *mcr*-2 in this study ($p = 0.217$ to 1.00). The results are detailed-out in Table 5.

4. Discussion

This study investigated phenotypic colistin-resistance and the carriage of *mcr*-1, *mcr*-2, *mcr*-3, *mcr*-4, and *mcr*-5 genes in

urinary tract infecting bacteria in Bangladesh. Our results found the emergence of colistin-resistant uropathogens carrying multiple *mcr* variants in Bangladesh. The findings attested to the earlier reports of *mcr* genes in some GNB isolated from clinical diseases in several other countries [34,35]; however, limited reports are available on *mcr*- associated UTI [21,36]. Our findings show harmony to the reported dispersion of *mcr* variants in clinical isolates of Bangladesh and Indian subcontinent countries [37–39]. The *mcr* genes were described primarily in farm-animal products [2,8,40] and thought to disseminate to humans rapidly by the zoonotic transmission [41,42]. High contents of *mcr* variants in different bacteria, isolated from poultry droppings, were recently reported from Bangladesh [13]. The findings identified food animals and the surrounding human-associated environments as noted reservoirs for the *mcr* genes to be transmitted into clinical isolates. Further reporting of colistin-resistant uropathogens with the *mcr* gene's presence in this study speculates a significant public health risk for colistin's clinical utility as a last resort drug. Being plasmid-mediated gene, *mcr* variants entail high mobility that can spread rapidly by vertical and horizontal gene transfer to other pathogenic and commensal bacteria [4,43]. Thereby, the *mcr* gene dissemination increases the public health burden of antimicrobial resistance and reduces the therapeutical effectiveness against multidrug-resistant bacteria.

Enterobacteriales was identified as a prevalent etiology of UTI in Bangladesh [44]. By two independent analyses, over 11% of the uropathogens carrying the hazardous *mcr*-1 and *mcr*-2 genes together with colistin-resistant phenotypes were detected. Our findings were congruent with the earlier reports of harboring *mcr* in the Enterobacteriale family [4,17,21,36]. 'ESKAPE pathogens,' *Pseudomonas* spp, *Klebsiella pneumoniae*, and *Enterobacter* spp

carrying *mcr-1* and *mcr-2* were identified [45–47]. Also, considerable proportions of *E. coli* in the study were found to carry the *mcr* variant, consistent with the earlier reports [8,43]. Acquisition of mobilized colistin resistance in the ‘ESKAPE pathogens’ may generate a new pathogenesis and transmission pattern [48].

Significantly higher MIC values for colistin for the *mcr*-positive isolates have been validated repeatedly about the association of colistin resistance and *mcr* genes in UTI isolates. Interestingly, some *mcr*-negative isolates (42.3%, 52/123) also showed phenotypic colistin-resistance by the broth dilution test. The genotype-phenotype association’s disagreement could be explained by other variants of the *mcr* gene, such as *mcr-6*, *mcr-7*, *mcr-8*, and *mcr-9*, that have not been investigated in this study [49,50]. The *mgrB* is a small transmembrane protein that helps binding polymyxin antibiotics (colistin and polymyxin B) to the gram-negative bacterial cell membrane. Insertional inactivation of the *mgrB* gene was associated with acquired colistin resistance in *Klebsiella pneumoniae* [51]. Similar mutational inactivation in the *mgrB* gene could further explain the colistin resistance in isolates that did not carry the *mcr* genes in this study. As colistin targets anionic lipid A phosphate moiety of bacterial LPS to cause membrane leakage, the *mcr*-negative bacteria further employ the modification phosphoethanolamine of lipid A in the cell membrane exhibit colistin resistance [52,53]. Activation of efflux pump was reported in addition to *mcr* gene to expel internalized polymyxin in *Klebsiella pneumoniae*, *Pseudomonas aeruginosa*, and in *Acinetobacter baumannii* [54–56].

Moreover, increased capsule polysaccharide production on the bacterial cell wall surface can provide polymyxin resistance [57,58]. Detailed studies will be required to determine the underlying mechanism for colistin resistance discrepancies without the *mcr* genes. All the phenotypic and genotypic colistin resistance results of this study have been assessed duplicates and are expected to be generalizable if similar studies to be conducted elsewhere in the country. The study findings carry public health importance in establishing potential surveillance and implementation of antimicrobial resistance stewardship in clinical settings in Bangladesh [59,60].

Females were diagnosed with UTI more commonly than males in this study. The female urinary system’s structural difference contributes acquisition of higher UTI [61]. Women in their reproductive years at the 20–39 age group accounted for the most UTI episodes in this study and elsewhere [62]. However, a higher proportion of the *mcr* carriage was noticed among male UTI patients (21.4%) than those of females (9.0%). The potentially important observation of gender-based dissimilar acquisition of the mobile colistin resistance genes needs further validation.

Phenotypic antibiotic susceptibility of *mcr*-positive isolates showed 85.6% resistance to Amoxicillin/clavulanic acid, and similar efficacy was observed for all generations of cephalosporin antibiotics. Some isolates exhibited susceptibility to the lower generation of cephalosporin but found resistance to their higher generation. A similar discrepancy was observed in aminoglycoside antibiotics as well. Further studies will be required to determine the phenotypic susceptibility to different generations of antibiotics. Elevated resistance to these antibiotics may be due to physicians’ empiric antimicrobial prescription, self-prescribing, non-adherence, and

over-the-counter drug consumption [63,64]. With the reduced potency of cephalosporins, clinicians have shifted prescribing more carbapenems. Although 7–40% of isolates were found resistant to imipenem, meropenem, or doripenem tested in this study. Previous studies conducted in Bangladesh identified carbapenem-resistant gram-negative UTI pathogens, indicating circulating carbapenem-resistance uropathogens in the country [65]. Again, the emergence of ESBL or carbapenemase-producing urinary pathogens can be linked with increased carbapenem consumption [66,67]. Altogether, more than 85% *mcr*-positive urinary tract pathogens were detected as MDR based on 17 antibiotics tested from six drug-groups. The emergence of the clinical MDR pathogens with resistance to the last resort reserved drug may cause untreatable infections and fatalities in the expanded population. The evolution analysis of *mcr* gene variants revealed their potential transmission from a focal point to all-over national and global distribution [4].

This study had several fundamental limitations. Risk behavior data of UTI was not collected in detail from the participants. The study was conducted through a cross-sectional design without prospective follow-up due to resource limitations. The recurrent infections were not separated from first-time infections. The purposive sampling was undertaken in some semi-urban populations with a notable degree of social stigma about urogenital diseases. Notably, women become uncomfortable visiting doctors/diagnostics for genitourinary diseases until the symptoms appear very severe. Therefore, the symptoms reported were more acute, and a higher proportion of positive UTI cases were identified than other related reports [68]. This study examined five *mcr* gene variants such as *mcr-1* to *mcr-5*; however, other recently reported variants, like, *mcr-6*, *mcr-7*, *mcr-8*, and *mcr-9*, were not investigated. The small sample size was another critical constraint to performing fully powered statistical analyses. However, our results were generated from a resource-limited setting and maintained internal validity by repeating independent experiments where necessary. Our findings have shed light on UTI pathogens’ composition, their carriage of *mcr* gene variants, their co-harborage, and genotypic-phenotypic association with colistin resistance. Further follow-up studies will assess the external validity of our findings.

5. Conclusions and recommendations

UTI remains a prevalent disease in Bangladesh. The mobile colistin resistance genes started emerging in Bangladeshi uropathogens. The *mcr-1* and *mcr-2* variants predominate in the Bangladeshi UTI bacteria over other variants, such as *mcr-3*, *mcr-4*, and *mcr-5*, which was found to have no association with either sex or age-groups. The presence of the *mcr* genes in UTI bacteria imposes significantly high phenotypic resistance to colistin, consequently endorsing more MDR uropathogens. The emergence of the clinical MDR pathogens with resistance to the last resort reserved drug may transmit untreatable infections and disease transmission into the unaffected cohorts. The findings entail urgent surveillance and intervention for national and global antimicrobial stewardship. As several reports found that colistin and other antibiotics are

being used as growth-promotor agents in animal husbandry in Bangladesh [13,69]. The aberrant uses of antibiotics in feeding animals increase antibiotic resistance transfer from animals to humans. Therefore, we strongly recommend immediate cessation of growth-promoting antibiotics use on livestock in Bangladesh.

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Declaration of interest

The authors have no relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript. This includes employment, consultancies, honoraria, stock ownership or options, expert testimony, grants or patents received or pending, or royalties.

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Author contributions

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ORCID

Bayasrin Ara  <http://orcid.org/0000-0003-2913-7863>
 Umme Laila Urmi  <http://orcid.org/0000-0003-1327-1638>
 Tanjum Ara Haque  <http://orcid.org/0000-0001-5850-7531>
 Shamsun Nahar  <http://orcid.org/0000-0002-3580-5576>
 Aditya Rumnaz  <http://orcid.org/0000-0001-5137-9944>
 Tamanna Ali  <http://orcid.org/0000-0001-9139-6095>
 Mohammed Shah Alam  <http://orcid.org/0000-0002-3238-2301>
 Abu Syed Md. Mosaddek  <http://orcid.org/0000-0002-7682-9187>
 nor Azlina a Rahman  <http://orcid.org/0000-0002-9046-6183>
 Mainul Haque  <http://orcid.org/0000-0002-6124-7993>
 Salequul Islam  <http://orcid.org/0000-0001-6131-4132>

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